EAST–WEST AND NORTH–SOUTH TRAFFIC

In network and service-architecture diagrams, traffic is often described by its direction. *North–south traffic* flows into or out of a system—from an external client or customer (north) down into the organization's infrastructure (south). These requests typically pass through edge components such as API gateways and firewalls that enforce perimeter security and validate OAuth tokens. At ACME, the API gateway mediates all north–south traffic, serving as the central enforcement point for external API access.

East–west traffic, on the other hand, moves within the system—between internal services, APIs, and databases. As ACME's Customer Collaboration platform evolved toward microservices, this internal traffic surged: a single customer request could trigger dozens of service-to-service calls. Traditional perimeter controls could no longer monitor or mediate these interactions. To secure this east–west flow, ACME implemented Cedar-based authorization in its service mesh and local, per-service proxies, ensuring that every internal call was evaluated against policy, just as external calls were at the gateway.

Relying solely on the gateway creates blind spots. If a bug or misconfiguration lets a service overreach, sensitive data could leak without detection. Bob suggested adopting a service mesh to fill this gap. With a mesh like Istio, each service-to-service call passes through a local sidecar proxy. These proxies establish mutual TLS (mTLS) to verify the identity of each service, and they can be configured to consult an external authorization system like Cedar before forwarding requests.

Alice's team integrated Cedar at this enforcement point. Each sidecar proxy became a policy enforcement point (PEP), sending a normalized PARC request to a Cedar engine as the policy decision point (PDP). Figure 11.2 illustrates how this worked when the Docs service sent content to the Search service.